

Practice Problem Set 1

ECON 300 – Labour Economics

Winter 2026

Problem 1: Labour Force Participation

In a city with a working-age population (15+) of 800,000:

- Employed: 520,000
- Unemployed: 40,000
- Not in the labour force: 240,000

1. Calculate the **labour force**.
2. Compute the **labour force participation rate (LFPR)**.
3. Compute the **unemployment rate**.
4. Compute the **employment-to-population ratio**.
5. Interpret the difference between the LFPR and the employment-to-population ratio.

Problem 2: Labour Supply and the Reservation Wage

A worker has the following weekly budget constraint:

$$C = 500 + 20h$$

where C is consumption (dollars) and h is hours worked. The worker has \$500 in non-labour income and earns \$20 per hour.

The worker's utility function is:

$$U = C - 0.5h^2$$

1. Derive the worker's **labour supply function**.
2. Find the optimal number of hours worked.
3. Determine the **reservation wage**.
4. Explain the economic meaning of the reservation wage.

Problem 3: Income and Substitution Effects

Suppose a worker earns \$25 per hour and works 40 hours per week. The wage increases to \$30 per hour.

1. Calculate the change in weekly income.
2. Using the labour–leisure model, explain:
 - the **substitution effect**, and
 - the **income effect**.
3. Under what conditions could hours worked fall after a wage increase?

Problem 4: Labour Supply and Leisure (Graphical Analysis)

Draw a standard **labour–leisure diagram** with leisure on the horizontal axis and consumption on the vertical axis.

1. Show the budget constraint for a wage of \$20 per hour.
2. Illustrate an increase in the wage to \$30 per hour.
3. Decompose the change in hours worked into:
 - substitution effect, and
 - income effect.
4. Label the **reservation wage** on the graph.

Problem 5: Market Labour Supply and Demand

Assume the labour market is described by:

$$L^D = 100 - 2w \quad L^S = 20 + 3w$$

where w is the wage.

1. Find the **equilibrium wage** and **employment**.
2. Graph the labour supply and demand curves.
3. Suppose a minimum wage of \$18 is imposed:
 - Is it binding?
 - Calculate unemployment.
4. Explain the result using economic intuition.